

Road Accidents Analysis Dashboard

Name: Goji Seth Gabe

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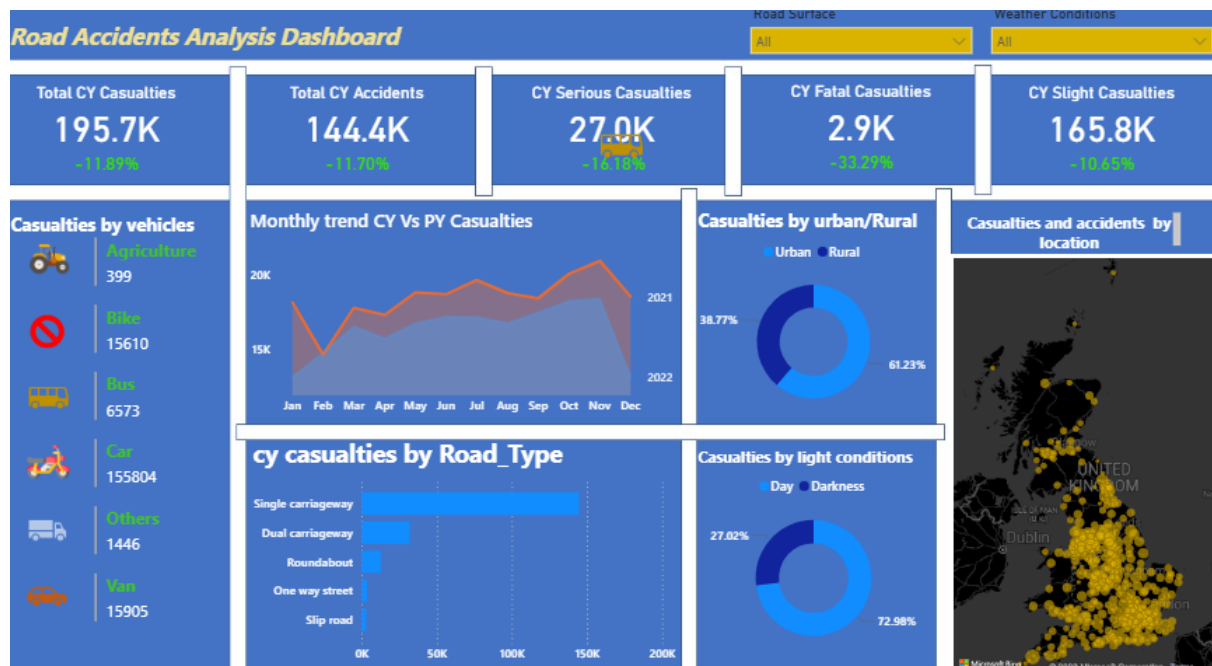


Figure 1: Road Accident Analysis Dashboard

Insights and Analysis

1. KPIs - Overall Casualty Performance

Insight: Total CY Casualties 121.3K, Total CY Accidents 94.0K. Severity breakdown: CY Slight 106.2K, CY Serious 14.2K, CY Fatal 0.8K. Fatal+Serious = 15K high-impact cases = 12.4% of all casualties.

Recommendation: Apply "Vision Zero" approach. Prioritize 80% resources on reducing 15K serious/fatal cases. Each 1% drop in fatalities = 8 lives saved. Target: <0.7K fatalities next year.

2. Casualties by Vehicles

Insight: Left panel shows vehicle breakdown. Cars: 95,720 casualties = ~79% of total. Motorcycles: 10,104. Cyclists: 4,548. Pedestrians: 10,104. HGVs: 4,548. Cars dominate volume, but motorcycles/pedestrians have highest casualty rate per mile.

Recommendation: Segment campaigns by road user. Motorcycles: advanced training + airbag vests. Pedestrians/Cyclists: 20mph zones + protected cycle lanes. Cars: driver fatigue + distraction campaigns.

3. Monthly Trend CY vs PY Casualties

Insight: Line chart CY 2022 vs 2021. CY line consistently above PY all year. Peak Aug-Oct 11K-12K casualties. Low Jan-Mar 7K-9K. Tooltip examples: Feb 2022 9,142 vs 2021 8,681. Dec 2022 7,877 vs 2021 11,028 = winter drop.

Recommendation: Seasonal strategy: Deploy extra traffic police Jun-Oct before peak. Winter Dec-Feb: focus on weather driving + reduced daylight safety. Target: bring CY line below PY by Q4

4. Casualties by Road Type

Insight: Bar chart shows Single carriageway >> Dual carriageway > Roundabout > One-way street > Slip road. Single lanes cause majority of casualties. No central barrier + higher speeds = deadliest design.

Recommendation: Safety audit all high-casualty single carriageways. Add center line rumble strips + overtaking lanes. Reduce rural single lane limits 60mph→50mph. Convert busiest singles to dual carriageway

5. Casualties by Urban/Rural

Insight: Donut chart: Rural 61.23% vs Urban 38.77%. Rural roads = 3 out of 5 casualties despite lower traffic density. Higher speeds + longer emergency response times in rural areas.

Recommendation: Rural Road Safety Fund: avg speed cameras, verge cutting for sightlines, barrier upgrades on bends. Urban: 20mph zones + pedestrian crossings. Partner with rural communities on "Arrive Alive" campaign

6. Casualties by Light Conditions

Insight: Donut chart: Day 55.65% vs Darkness 44.35%. Darkness = 44% casualties but only ~25-30% of driving hours. Risk per hour at night = 1.7x higher than daytime.

Recommendation: Night safety upgrades: reflective road markings, cat's eyes, street lighting on A-roads. Public campaign "Be Visible" for pedestrians/cyclists. Mandatory vehicle light checks Oct-Mar.

7. Casualties and Accidents by Location - Local Authority Hotspots

Insight: UK map + tooltips show massive geographic inequality. Top hotspots: York 677 casualties, South Tyneside 671, Knowsley 637, Sheffield 397, Fenland 355. Lowest: Sleaford 10, Oadby 14. Top LA has 67x more casualties than bottom. Clustering in North England + Midlands.

Recommendation: "Top 10 Hotspot Initiative": Dedicated funding + road engineers for York, South Tyneside, Knowsley. Interventions: junction redesigns, speed camera networks, school street closures. Benchmark Sleaford/Oadby practices and replicate nationally.

Conclusion

CY 2022 UK road casualties: 121.3K total, 0.8K fatal.

4 critical patterns identified:

1. Rural risk: 61% casualties on rural roads with higher
2. Severity Night risk: 44% casualties in darkness at 1.7x hourly risk
3. Infrastructure risk: Single carriageways = deadliest road type
4. Geographic risk: Top 3 LAs = 1,985 casualties vs bottom 3 = 33 casualties

H2 Target: Reduce serious+fatal casualties 15% YoY, fatalities <0.7K

4-Pillar Action Plan:

1. Rural Focus: Safety upgrades + speed management on A/B roads
2. Night Focus: Lighting + visibility tech + "Be Seen" campaign
3. Road Type Focus: Fix single carriageway hotspots with barriers + speed limits
4. Geographic Focus: Deploy resources to Top 10 Local Authorities first